

Q1. Quadratic equation

Score: 10.00 TC: 2/2 passed

CODE

```
#include <stdio.h>
int main()
{
    float a, b, c, d, root1, root2;
    scanf("%f %f %f", &a, &b, &c);
    d = b*b - 4*a*c;
    if (d > 0){
        root1 = (-b + sqrt(d)) / (2 * a);
        root2 = (-b - sqrt(d)) / (2 * a);
        printf("%.2f %.2f\n", root1, root2);
    }
    return 0;
}
```

INPUT

1 -5 6

OUTPUT

3.00 2.00

Q2. Prime numbers

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>
int main()
{
    int n, i, j, prime, count = 0;
    if(scanf("%d", &n) != 1){
        return 0;
    }
    for (i = 2; i ≤ n; i++) {
        prime = 1;
        for (j = 2; j ≤ i / 2; j++) {
            if (i % j == 0){
                prime = 0;
                break;
            }
        }
        if (prime){
            printf("%d ", i);
            count++;
        }
    }
    if (count == 0){
        printf("No prime numbers");
    }
    return 0;
}
```

INPUT

2

OUTPUT

2

Q3. Palindrome**Score: 10.00** **TC: 3/3 passed****CODE**

```
#include <stdio.h>
int main() {
    char s[100];
    int i = 0, l = 0, p = 1;
    scanf("%99s", s);
    for(i = 0; s[l] != '\0'; i++)
        l++;
    for (i = 0; i ≤ l/2; i++) {
        if(s[i] != s[l-1-i]) {
            p = 0;
            break;
        }
    }
    if(p)
        printf("Palindrome");
    else
        printf("Not a Palindrome");
    return 0;
}
```

INPUT

hello

OUTPUT

Not a Palindrome

Q4. Pyramid**Score: 10.00** **TC: 3/3 passed****CODE**

```
#include <stdio.h>
int main()
{
    int n, i, j, space;
    if(scanf("%d", &n) != 1){
        return 0;
    }
    for (i = 1; i ≤ n; i++) {
        for (space = 1; space ≤ n - i; space++){
            printf(" ");
        }
        for (j = 1; j ≤ i; j++){
            printf("%d", j);
        }
        for (j = i - 1; j ≥ 1; j--){
            printf("%d", j);
        }
        printf("\n");
    }
    return 0;
}
```

INPUT

3

OUTPUT1
121
12321

Q5. Voting**Score: 10.00** **TC: 2/2 passed****CODE**

```
#include <stdio.h>
int main() {
    int age;
    scanf("%d", &age);
    if (age ≥ 18)
        printf("Eligible");
    else
        printf("Not eligible");
    return 0;
}
```

INPUT

18

OUTPUT

Eligible

Q6. Calculator**Score: 0.00** **TC: 0/3 passed****CODE**

```
#include <stdio.h>
int main() {
    int a, b;
    char op;
    scanf("%d %c %d", &a, &op, &b);
    switch (op) {
        case '+':
            printf("Add result = %d", a + b);
            break;
        case '-':
            printf("Subtrsrction result = %d", a - b);
            break;
        case '*':
            printf("Multiplication = %d", a * b);
            break;
        case '/':
            if (b ≠ 0)
                printf("Error: Division by zero");
            else
                printf("Division result = %d", a / b);
            break;
        default:
            printf("Invalid operator");
    }
    return 0;
}
```

INPUT

8 * 7

OUTPUT

Multiplication = 56

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Q7. Month counting

Score: 0.00 TC: 0/2 passed

CODE

```
#include <stdio.h>
int main() {
    int month;
    if(scanf("%d", &month) != 1){
        return 0;
    }
    printf("Input month number:%d\n", month);
    switch (month) {
        case 1: case 3: case 5: case 7:
        case 8: case 10: case 12:
            printf("Output:31");
            break;
        case 4: case 6: case 9: case 11:
            printf("Output:30");
            break;
        case 2:
            printf("Output:28");
            break;
        default:
            printf("Invalid month");
    }
    return 0;
}
```

INPUT

2

OUTPUT

Input month number:2
Output:28

Q8. Positive or negative

Score: 10.00 TC: 2/2 passed

CODE

```
#include <stdio.h>
int main() {
    int num;
    scanf("%d", &num);
    if (num > 0)
        printf("Positive");
    else if (num < 0)
        printf("Negative");
    else
        printf("Zero");
    return 0;
}
```

INPUT

-8

OUTPUT

Negative

Q9. Multiplication Table**Score: 10.00** **TC: 2/2 passed****CODE**

```
#include <stdio.h>
int main() {
    int num, i;
    if(scanf("%d", &num) != 1){
        return 0;
    }
    for (i = 1; i ≤ 10; i++) {
        printf("%d x %d = %d\n", num, i, num * i);
    }
    return 0;
}
```

INPUT

12

OUTPUT

```
12 x 1 = 12
12 x 2 = 24
12 x 3 = 36
12 x 4 = 48
12 x 5 = 60
12 x 6 = 72
12 x 7 = 84
12 x 8 = 96
12 x 9 = 108
12 x 10 = 120
```

Q10. Factorial**Score: 10.00** **TC: 3/3 passed****CODE**

```
#include <stdio.h>
int main() {
    int n, i;
    long long fact = 1;
    scanf("%d", &n);
    for(i = 1; i ≤ n; i++){
        fact = fact * i;
    }
    printf("%lld", fact);
    return 0;
}
```

INPUT

5

OUTPUT

120